V.L Memorial Public School

475 Vijay Nagar

CHAPTER-14-QUESTIONS-CLASS7TH-MATHS

Class 07 - Maths (Ganita Prakash)

Section A

1. A 9×12 rectangular grid is to be tiled using 2×1 domino tiles. Which statement is correct? **[1]**

   a) It is tileable because 9×12 is even

   b) It is not tileable because 12 is even

   c) It is tileable only if one corner square is removed

   d) It is not tileable because 9 is odd

2. What condition must be met for two arcs to create a symmetric eye? **[1]**

   a) They must have equal distances from a baseline

   b) They must intersect at a point

   c) They must have different radii

   d) They must be drawn using the same center

3. To construct a 105° angle using a compass, which two angles must be combined or bisected? **[1]**

   a) 60° and 45°

   b) 90° and 120°

   c) All of these

   d) 90° and 15°

4. You construct a 90° angle at a point O on a line l by first making O the midpoint of XY on l and then drawing the perpendicular bisector of XY. **[1]**
   Which idea is being used in this construction?

   a) Exterior angle property of triangles

   b) Any point on a perpendicular bisector is equidistant from the endpoints

   c) Midpoint theorem of triangles

   d) Sum of angles in a triangle is 180°

5. In the context of geometric constructions, what does symmetry refer to? **[1]**

   a) The alignment of angles

   b) The equal distribution of weight

   c) The balance and harmony of shapes

   d) The proportionality of elements

6. What is one practical application of geometric constructions in everyday life? **[1]**

   a) Architecture

   b) Gardening

   c) Writing

   d) Cooking

7. The number of V-shaped angles used to form a Kanizsa triangle is **[1]**

   a) 4

   b) 3

   c) None of these

   d) 2

8. How can geometric constructions influence art and design? **[1]**

a) They complicate the design process
b) They provide structure and precision
c) They emphasize randomness
d) They limit creativity

9. Which geometric shape has all sides equal and is commonly used in rangoli symmetry patterns? **[1]**

a) Oval
b) Rectangle
c) Triangle
d) Square

10. What mathematical principle is essential for creating a perfectly symmetrical design? **[1]**

a) The Pythagorean theorem
b) Proportional relationships
c) The Golden Ratio
d) Equal dimensions

11. To divide a line segment into 16 equal parts, the total number of perpendicular bisectors that need to be drawn is **[1]**

a) 4
b) 16
c) 15
d) 8

12. What is the purpose of the supporting line XY in the construction of eyes? **[1]**

a) To measure the length of the eyes
b) To ensure symmetry
c) To create a straight edge
d) To mark the center

13. Which of the following regular polygons alone can tile the entire plane without gaps or overlaps? **[1]**

a) All regular polygons
b) Only equilateral triangles and squares
c) Only regular pentagons and hexagons
d) Equilateral triangles, squares and regular hexagons

14. To tile an $m \times n$ grid using 2×1 tiles, which of the following must be true? **[1]**

a) $m \times n$ must be odd.
b) m must be odd.
c) $m \times n$ must be even.
d) n must be odd.

15. What mathematical concept is prominently featured in the design of honeycombs by bees? **[1]**

a) Circles
b) Squares
c) Hexagons
d) Triangles

16. A hexagon is called a regular hexagon if: **[1]**

a) all sides and angles are equal
b) either sides or angles, but not both
c) all sides are equal
d) all angles are equal

17. Why do the upper and lower arcs of the eye need to have the same radius? **[1]**

a) To fit within a specific shape
b) To maintain symmetry
c) For aesthetic purposes
d) To ensure they touch at the baseline

18. In geometric constructions, how can one ensure that both eyes look balanced? **[1]**

a) By matching the arcs precisely
b) By using different colors
c) By adjusting the size randomly
d) By measuring with a ruler

19. Which of the following best describes the role of arcs in geometric constructions? **[1]**

a) They serve as measurement tools
b) They create sharp angles
c) They form straight lines
d) They allow for smooth curves

20. How does understanding geometric constructions enhance problem-solving skills? **[1]**

a) By increasing memorization only
b) By simplifying all calculations automatically
c) By providing rigid guidelines
d) By fostering analytical skills

21. **Assertion (A):** A 60° angle can be constructed by constructing an equilateral triangle. **[1]**
Reason (R): All sides of an equilateral triangle are equal.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** An angle bisector divides an angle into two equal parts. **[1]**
Reason (R) An angle bisector is always perpendicular to the arms of the angle.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** A 5 × 7 grid can be tiled using 2 × 1 tiles. **[1]**
Reason (R): Each 2 × 1 tile covers exactly two unit squares.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Any point on the perpendicular bisector of a line segment is equidistant from its endpoints. **[1]**
Reason (R): The perpendicular bisector divides a line segment into two equal parts at right angles.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** Equilateral triangles can tile the entire plane without gaps or overlaps. **[1]**
Reason (R): The sum of interior angles of an equilateral triangle is 180°.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. If a point A is such that AX = AY, where X and Y are endpoints of a segment, on which line does A lie? **[1]**

27. How many equal parts does an angle bisector divide an angle into? **[1]**

28. In a black-white coloured grid, how many black and white squares must a 2 × 1 tile cover? **[1]**

29. A region has 14 black squares and 12 white squares. Can it be tiled using 2 × 1 tiles? **[1]**

30. Can a 5 × 7 grid be tiled using 2 × 1 tiles? Justify. **[1]**

31. How many degrees are there in each angle when a full circle is divided into 8 equal parts? **[1]**

32. Can a perpendicular bisector of a line segment be drawn by using only a ruler? **[1]**

33. When constructing the perpendicular bisector, is it necessary to have the same radius of the arcs above and below the line? **[1]**

34. How many equilateral triangles meet at a point when they tile the plane? **[1]**

35. If a point on the perpendicular bisector is joined to the ends of the line segment, what type of triangle is formed? **[1]**

36. What is the angle between a line and its perpendicular bisector? **[1]**

37. What is the measure of each angle of an equilateral triangle? **[1]**

Section B

38. In the construction of a symmetric eye, why must the two arcs have equal radius and what happens to the figure if the radii are different? **[2]**

39. How many degrees is each angle of an equilateral triangle and how is this fact used to construct a 60° angle? **[2]**

40. While copying an angle using ruler and compass, which triangle congruence rule is used and why does this rule guarantee the copied angle is equal to the given angle? **[2]**

41. How many equilateral triangles meet at one point in a regular triangular tiling and find the total angle formed at that point. **[2]**

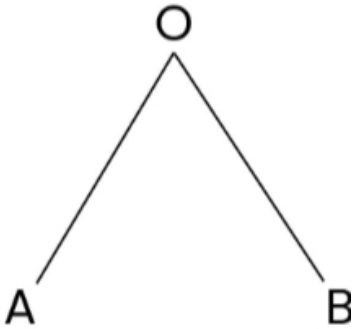
42. A full circle is used to draw an 8-petalled design; find the measure of each central angle and also state which construction idea from the chapter is used to obtain this angle. **[2]**

43. Name one regular polygon other than a square that can tile the plane and also state the angle condition that makes this tiling possible. **[2]**

44. Find the total number of unit squares in a 6 × 9 grid and state whether it can be tiled using 2 × 1 tiles. **[2]**

Section C

45. An architect wants to draw a perfectly symmetrical pointed arch for a gateway. She first draws two line segments OA and OB of equal length meeting at O at the top, forming a "V" shape. **[3]**



a. Explain why OA and OB must be equal for the arch to look symmetrical.

b. Describe in words how she can use a compass to turn this "V" shape into a smooth pointed arch.

46. A school wants to design an 8-petalled rangoli pattern in its courtyard using chalk. They will draw supporting lines first and then draw arcs. **[3]**

a. How many 90° angles do they need to construct to get 8 equal sectors?

b. Explain step by step how they can obtain all eight 45° supporting lines using only constructions.

47. A hall floor is to be tiled entirely using regular polygons of just one type. The engineer suggests three options: **[3]**

- Option 1: Regular pentagons
- Option 2: Regular hexagons
- Option 3: A mix of regular pentagons and regular hexagons

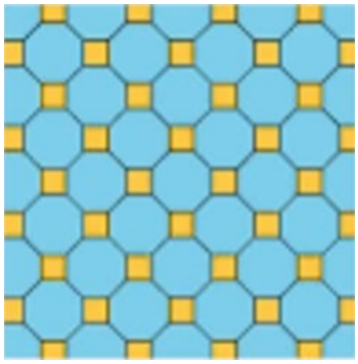
a. Which option(s) can tile the hall without gaps or overlaps if only one type of regular polygon is allowed?

b. Give a mathematical reason for your answer based on interior angles and tiling.

48. Explain how the bisection of an angle is used to divide a full circle into equal parts. **[3]**

49. Explain the concept of angle bisection. How is it useful in creating geometric patterns? **[3]**

50. A polygon tiles the plane with no gaps. Three such polygons meet at a point. Each polygon has an interior angle of 135°. Is such a tiling possible? Justify mathematically. **[3]**



51. How do you construct the perpendicular bisector of a line segment using a compass and ruler? **[3]**

52. A 7 × 7 board has one corner square removed. A student claims: **[3]**

"The board now has 48 squares, which is even, so it must be possible to tile it with 2 × 1 tiles."

a. Is the student's reasoning correct?

b. Justify your answer using the checkerboard colouring idea.

Section D

53. **Read the following text carefully and answer the questions that follow:** **[4]**

Pointed arches are architectural elements featuring two curves meeting at a sharp top angle.

Answer the following questions based on the image of a pointed arc provided below:



Questions:

a. How is a pointed arch different from a semicircular arch? **(1)**

b. Where are pointed arches commonly seen in buildings or structures? **(1)**

c. Explain how a pointed arch is formed. **(2)**

OR

If their midpoints are marked, will you be able to construct a pointed arch? **(2)**

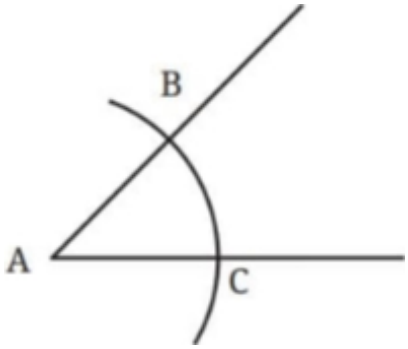
Section E

54. A triangle ABC is such that AB = AC. A student constructs the perpendicular bisector of BC and claims that it passes through A even if AB ≠ AC. Is the claim correct? Explain using geometric reasoning. **[5]**

55. A teacher copies a 60° angle using compass-and-arc construction. A student copies that copied angle, another student copies the student's copy, and so on for 12 students. If each copying introduces a tiny 0.5° error, will the final angle still be accurate enough to tile around a point? Explain. **[5]**

56. Construct at least 4 different angles. Draw their bisectors. **[5]**

57. You are asked to copy an angle ∠A using only ruler and compass. During construction, the arc from A intersects the two arms at B and C. A student mistakenly measures AC instead of BC and transfers this length to the new arc. Will the copied angle still be correct? Justify your answer. **[5]**



58. How do we construct a 60° angle? **[5]**

59. Explain the process of creating a regular tiling pattern using congruent triangles. Describe the steps and explain how symmetry plays a key role in creating the pattern. **[5]**

60. For a line segment XY of length 8 cm, two points A and B are chosen such that AX = AY = BX = BY. Without drawing, determine whether line AB will still bisect XY if X and Y are moved farther apart, but A and B remain equidistant from both X and Y. Explain with geometric reasoning. **[5]**

61. A designer wants to draw two perfectly identical "leaf-shaped eyes". She marks a line XY and chooses two different points P and Q above it such that PX = PY and QX = QY. Explain why her design will always stay symmetric. **[5]**

Section F

62. **Fill in the blanks:** **[6]**

(a) A region can be tiled with 2 × 1 tiles only if the number of black and white squares are _____. **[1]**

(b) To construct a regular hexagon inside a circle, the central angle between any two adjacent vertices should be _____ degree angle with the original line. **[1]**

(c) The line that divides a line segment into two equal parts and is perpendicular to it is called the _____. **[1]**

(d) If an angle is bisected to form two equal angles 68°, then the measure of the original **[1]**

angle is _____.

(e) An angle bisector divides an angle into _____ equal parts. **[1]**

(f) A rectangular grid with an odd number of unit squares _____ be tiled using 2×1 tiles. **[1]**